

$$A_1 = \begin{pmatrix} 1 & 3 & 4 \\ 2 & 1 & 3 \\ 4 & 2 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 3 & 4 & | & 1 & 0 & 0 \\ 2 & 1 & 3 & | & 0 & 1 & 0 \\ 4 & 2 & 1 & | & 0 & 0 & 1 \end{pmatrix} \quad \begin{array}{l} F_2 - 2F_1 \rightarrow F_2 \\ F_3 - 4F_1 \rightarrow F_3 \end{array}$$

$$\begin{pmatrix} 1 & 3 & 4 & | & 1 & 0 & 0 \\ 0 & -5 & -5 & | & -2 & 1 & 0 \\ 0 & -10 & -15 & | & -4 & 0 & 1 \end{pmatrix} \xrightarrow{-1/5 F_2 + F_3} \begin{pmatrix} 1 & 3 & 4 & | & 1 & 0 & 0 \\ 0 & -5 & -5 & | & -2 & 1 & 0 \\ 0 & -10 & -15 & | & -4 & 0 & 1 \end{pmatrix} \quad \begin{array}{l} F_2 + F_3 - 5F_2 \\ F_3 + F_3 + 10F_2 \end{array}$$

$$\begin{pmatrix} 1 & 0 & 7 & | & -1/5 & 1/5 & 0 \\ 0 & 1 & 1 & | & 2/5 & -1/5 & 0 \\ 0 & 0 & -5 & | & 0 & -2 & 1 \end{pmatrix} \xrightarrow{F_1 + 1/5 F_3} \begin{pmatrix} 1 & 0 & 7 & | & -1/5 & 3/5 & 0 \\ 0 & 1 & 1 & | & 2/5 & -1/5 & 0 \\ 0 & 0 & -5 & | & 0 & -2 & 1 \end{pmatrix}$$

$$\begin{array}{l} F_1 \leftarrow F_1 - F_3 \\ F_2 \leftarrow F_2 - F_3 \end{array} \begin{pmatrix} 1 & 0 & 0 & | & -1/5 & 1/5 & 1/5 \\ 0 & 1 & 0 & | & 2/5 & -3/5 & 1/5 \\ 0 & 0 & 1 & | & 0 & 2/5 & -1/5 \end{pmatrix} \quad A_1^{-1} = \begin{pmatrix} -1/5 & 1/5 & 1/5 \\ 2/5 & -3/5 & 1/5 \\ 0 & 2/5 & -1/5 \end{pmatrix}$$

$$A_2 = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 4 & | & 0 & 1 & 0 \\ 5 & 6 & 0 & | & 0 & 0 & 1 \end{pmatrix} \quad F_3 - 5F_1 + F_3 \quad \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 4 & | & 0 & 1 & 0 \\ 0 & -4 & -15 & | & -5 & 0 & 1 \end{pmatrix}$$

$$\begin{array}{l} F_1 - 2F_2 + F_1 \\ F_3 + 4F_2 \rightarrow F_3 \end{array} \begin{pmatrix} 1 & 0 & -5 & | & 1 & -2 & 0 \\ 0 & 1 & 4 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & -5 & 4 & 1 \end{pmatrix} \quad \begin{array}{l} F_2 - 4F_3 \rightarrow F_2 \\ F_1 + 5F_3 \rightarrow F_1 \end{array} \begin{pmatrix} 1 & 0 & 0 & | & -24 & 18 & 5 \\ 0 & 1 & 0 & | & 20 & -15 & -4 \\ 0 & 0 & 1 & | & -5 & 4 & 1 \end{pmatrix}$$

$$A_2^{-1} = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix} //$$

$$A_3 = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 3 & 4 \end{pmatrix} \quad \begin{pmatrix} 4 & 2 & 1 & | & 1 & 0 & 0 \\ 2 & 1 & 3 & | & 0 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{pmatrix} \quad F_1 \leftarrow 1/4 F_1 \quad \begin{pmatrix} 1 & 1/2 & 1/4 & | & 1/4 & 0 & 0 \\ 2 & 1 & 3 & | & 0 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{array}{l} F_2 \leftarrow F_2 - 2F_1 \\ F_3 \leftarrow F_3 - F_1 \end{array} \begin{pmatrix} 1 & 1/2 & 1/4 & | & 1/4 & 0 & 0 \\ 0 & 0 & 5/2 & | & -1/2 & 1 & 0 \\ 0 & 5/2 & 15/4 & | & -3/4 & 0 & 1 \end{pmatrix} \quad F_2 \leftrightarrow F_3 \quad \begin{pmatrix} 1 & 1/2 & 1/4 & | & 1/4 & 0 & 0 \\ 0 & 5/2 & 15/4 & | & -3/4 & 0 & 1 \\ 0 & 0 & 5/2 & | & -1/2 & 1 & 0 \end{pmatrix}$$

$$F_2 \leftarrow 2/5 F_2 \quad \begin{pmatrix} 1 & 1/2 & 1/4 & | & 1/4 & 0 & 0 \\ 0 & 1 & 3/2 & | & -1/10 & 2 & 2/5 \\ 0 & 0 & 5/2 & | & -1/2 & 1 & 0 \end{pmatrix} \quad F_1 \leftarrow F_1 - 1/2 F_2 \quad \begin{pmatrix} 1 & 0 & -1/2 & | & 3/10 & -1 & -1/5 \\ 0 & 1 & 3/2 & | & -1/10 & 2 & 2/5 \\ 0 & 0 & 5/2 & | & -1/2 & 1 & 0 \end{pmatrix}$$

$$F_3 \leftarrow 2/5 F_3 \quad \begin{pmatrix} 1 & 0 & -1/2 & | & 3/10 & -1 & -1/5 \\ 0 & 1 & 3/2 & | & -1/10 & 2 & 2/5 \\ 0 & 0 & 1 & | & -1/5 & 2/5 & 0 \end{pmatrix} \quad \begin{array}{l} F_1 \leftarrow F_1 + 1/2 F_3 \\ F_2 \leftarrow F_2 - 3/2 F_3 \end{array} \begin{pmatrix} 1 & 0 & 0 & | & 1/5 & -3/5 & 1/5 \\ 0 & 1 & 0 & | & 1/5 & 8/5 & 2/5 \\ 0 & 0 & 1 & | & -1/5 & 2/5 & 0 \end{pmatrix}$$

$$A_1 = \left(\begin{array}{cccc|cccc} 2 & 4 & 6 & 1 & 1 & 0 & 0 & 0 \\ 4 & 7 & 5 & -6 & 0 & 1 & 0 & 0 \\ 2 & 5 & 18 & 10 & 0 & 0 & 1 & 0 \\ 6 & 12 & 38 & 16 & 0 & 0 & 0 & 1 \end{array} \right) \quad F_1 \leftarrow \frac{1}{2} F_1 \quad \left(\begin{array}{cccc|cccc} 1 & 2 & 3 & 1/2 & 1/2 & 0 & 0 & 0 \\ 4 & 7 & 5 & -6 & 0 & 1 & 0 & 0 \\ 2 & 5 & 18 & 10 & 0 & 0 & 1 & 0 \\ 6 & 12 & 38 & 16 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{aligned} F_2 &\leftarrow F_2 - 4F_1 \\ F_3 &\leftarrow F_3 - 2F_1 \\ F_4 &\leftarrow F_4 - 6F_1 \end{aligned} \quad \left(\begin{array}{cccc|cccc} 1 & 2 & 3 & 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 3 & -7 & -8 & 2 & -1 & 0 & 0 \\ 0 & 1 & 12 & 9 & -1 & 0 & 1 & 0 \\ 0 & 0 & 20 & 13 & -3 & 0 & 0 & 1 \end{array} \right) \quad F_2 \leftrightarrow F_3 \quad \left(\begin{array}{cccc|cccc} 1 & 2 & 3 & 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 1 & 12 & 9 & -1 & 0 & 1 & 0 \\ 0 & 3 & -7 & -8 & 2 & -1 & 0 & 0 \\ 0 & 0 & 20 & 13 & -3 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{aligned} F_1 &\leftarrow F_1 - 2F_2 \\ F_3 &\leftarrow F_3 - F_2 \end{aligned} \quad \left(\begin{array}{cccc|cccc} 1 & 0 & -11 & -17/2 & -1/2 & 2 & 0 & 0 \\ 0 & 1 & 12 & 9 & -1 & 0 & 1 & 0 \\ 0 & 0 & 5 & 1 & -3 & 1 & 1 & 0 \\ 0 & 0 & 20 & 13 & -3 & 0 & 0 & 1 \end{array} \right) \quad F_3 \leftarrow \frac{1}{5} F_3 \quad \left(\begin{array}{cccc|cccc} 1 & 0 & -11 & -17/2 & -1/2 & 2 & 0 & 0 \\ 0 & 1 & 12 & 9 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1/5 & -3/5 & 1/5 & 1/5 & 0 \\ 0 & 0 & 20 & 13 & -3 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{aligned} F_1 &\leftarrow F_1 + 11F_3 \\ F_2 &\leftarrow F_2 - 12F_3 \\ F_4 &\leftarrow F_4 - 20F_3 \end{aligned} \quad \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & -131/10 & 27/5 & 11/5 & 0 & 0 \\ 0 & 1 & 0 & 23/5 & -12/5 & -3/5 & 0 & 0 \\ 0 & 0 & 1 & 1/5 & 1/5 & 1/5 & 0 & 0 \\ 0 & 0 & 0 & 9 & -4 & -4 & 1 & 1 \end{array} \right)$$

$$A_1^{-1} = \left(\begin{array}{cccc} -11/10 & 4/5 & 3/5 & 13/90 \\ 3/5 & -1/5 & -4/5 & 11/45 \\ -3/5 & 1/5 & 1/5 & -1/45 \\ 1 & -4/9 & -4/9 & 1/9 \end{array} \right)$$

Descomposición LU

$$A_1 = \left(\begin{array}{ccc} 1 & 3 & 4 \\ 2 & 1 & 3 \\ 4 & 2 & 1 \end{array} \right) \quad \begin{aligned} m_{21} &= 2 & F_2 &\leftarrow F_2 - 2F_1 \\ m_{31} &= 4 & F_3 &\leftarrow F_3 - 4F_1 \end{aligned} \quad \left(\begin{array}{ccc} 1 & 3 & 4 \\ 0 & -5 & -5 \\ 0 & -10 & -15 \end{array} \right) \quad \begin{aligned} m_{32} &= \frac{-10}{-5} = 2 \\ & & & \end{aligned}$$

$$L = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 2 & 1 \end{array} \right) \quad U = \left(\begin{array}{ccc} 1 & 3 & 4 \\ 0 & -5 & -5 \\ 0 & 0 & -5 \end{array} \right)$$

$$A_2 = \left(\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{array} \right) \quad \begin{aligned} m_{31} &= 5 & F_3 &\leftarrow F_3 - 5F_1 \\ m_{32} &= -9 & F_3 &\leftarrow F_3 + 9F_2 \end{aligned} \quad \left(\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & -4 & -15 \end{array} \right)$$

$$L = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & -4 & 1 \end{array} \right) \quad U = \left(\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{array} \right)$$

$$A_3 = \begin{pmatrix} 4 & 2 & 7 \\ 2 & 1 & 3 \\ 7 & 3 & 4 \end{pmatrix} \quad F_2 \leftrightarrow F_3 \quad \begin{pmatrix} 4 & 2 & 7 \\ 17 & 3 & 4 \\ 2 & 1 & 3 \end{pmatrix} \quad m_{21} = 1/4 \quad F_2 \leftarrow F_2 - 1/4 F_1$$

$$m_{31} = 1/4 = 7/4 \quad F_3 \leftarrow F_3 - 7/4 F_1$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1/4 & 1 & 0 \\ 1/2 & 2/5 & 1 \end{pmatrix} //$$

$$U = \begin{pmatrix} 4 & 2 & 7 \\ 0 & 5/2 & 15/4 \\ 0 & 0 & 5/2 \end{pmatrix} //$$